Inverse Edge Mostar Indices of c-Cyclic Molecular Graphs

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Abstract

The edge Mostar index is a distance-based bond-additive invariant that measures, for every bond of a molecular graph, the imbalance between the numbers of bonds closer to its two end vertices. Motivated by inverse problems for topological indices in chemical graph theory, Alex and Gutman asked whether, for every fixed cyclomatic number $c \geq 3$, all sufficiently large positive integers occur as edge Mostar indices of c-cyclic graphs. We answer this question in the affirmative, and we do so inside the usual class of molecular graphs, namely simple connected graphs of maximum degree at most four. For every fixed $c \geq 3$ we construct explicit one-parameter families whose cyclomatic number is c. The construction is based on a long cycle with two short chords and a prescribed number of two-edge ears placed in two separated ear regions. A complete edge-distance classification is given: for each parity case, every edge is assigned its exact edge-Mostar contribution. Consequently the edge Mostar index is an affine function of the length of the long cycle in each parity class, and the two parity classes cover all sufficiently large integers. Explicit finite exceptional sets are obtained, although no minimality is claimed.

Keywords: edge Mostar index; molecular graph; inverse problem; cyclomatic number; chemical graph theory.
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1 Introduction

Molecular graphs are among the classical mathematical models used in chemical graph theory. In the hydrogen-suppressed model, vertices represent atoms of the molecular skeleton and edges represent covalent bonds. In particular, carbon skeletons are naturally modeled by simple connected graphs of maximum degree at most four, reflecting tetravalence; this convention is standard in the theory of molecular descriptors and topological indices [2, 3, 4]. In this paper a molecular graph means a simple connected graph G with $\Delta(G) \leq 4$.

For a connected graph $G = (V(G), E(G))$, the cyclomatic number is

$$c(G) = |E(G)| - |V(G)| + 1.$$

It is the number of independent cycles of G. In the language of chemical graph theory, this parameter gives a graph-theoretic measure of ring complexity: $c(G) = 0$ corresponds to a tree-like skeleton, $c(G) = 1$ to a unicyclic skeleton, and larger values to polycyclic systems.

Distance-based topological indices have been used since Wiener's pioneering study of paraffin boiling points [1]. The Mostar index was introduced by Došlić et al. as a measure of distance imbalance of edges [5]. It has since been studied both as a graph invariant and as a molecular descriptor; Mostar-type indices have been computed for carbon nanostructures and benzenoid systems [6], and a survey is given in [7]. The edge Mostar index, which counts closer edges rather than closer vertices, appears naturally as an edge analogue of this imbalance concept [6, 8].

We use the following notation throughout the paper. If x is a vertex and $f = ab$ is an edge of G, set

$$\operatorname{dist}_G(x, f) = \min\{\operatorname{dist}_G(x, a), \operatorname{dist}_G(x, b)\},$$

where dist_G denotes the ordinary shortest-path distance in G. For an edge $e = uv \in E(G)$, let

$$m_u(e \mid G) = \big|\{f \in E(G) : \operatorname{dist}_G(u, f) < \operatorname{dist}_G(v, f)\}\big|,$$

and define $m_v(e \mid G)$ analogously. The contribution of $e = uv$ to the edge Mostar index is

$$\mu_G(e) = \big|m_u(e \mid G) - m_v(e \mid G)\big|,$$

and

$$\mathrm{Mo}_e(G) = \sum_{e \in E(G)} \mu_G(e).$$

When the graph is clear from context, we simply write $\mu(e)$.

The inverse problem asks which integers occur as values of a given topological index on a prescribed graph class. Alex and Gutman settled several inverse Mostar and inverse edge Mostar questions for unicyclic and bicyclic molecular graphs and proposed the following higher-cyclomatic-number problem [12].

Conjecture 4.2. For any fixed $c \geq 3$, there exists a finite set $B \subset \mathbb{N}$ such that every $p \in \mathbb{N} \setminus B$ is the edge Mostar index of a c-cyclic graph.

Here and below $\mathbb{N} = \{1, 2, 3, \ldots\}$, and a graph is called c-cyclic if its cyclomatic number is exactly c. The main result of this paper proves the conjecture, in fact in the stronger molecular-graph setting.

2 The graph family and the ear regions

Let

$$C_L = v_0 v_1 \cdots v_{L-1} v_0$$

be a cycle of length L. Subscripts on the vertices of C_L are read modulo L only when this is explicitly stated. We write

$$C_i = v_i v_{i+1} \quad (0 \leq i \leq L-2), \qquad C_{L-1} = v_{L-1} v_0$$

for the cycle edges.

We now define a graph $G_{a,b}(L)$, where $a \geq 0$ and $b \geq 1$ are integers. Start with C_L and add two chords

$$A = v_{L-1} v_1, \qquad B = v_0 v_2.$$

Let

$$s = \left\lfloor \frac{L}{2} \right\rfloor.$$

Add new vertices

$$x_0, x_1, \ldots, x_{a-1}, \qquad y_0, y_1, \ldots, y_{b-1}.$$

For $0 \le i \le a-1$, add the two edges

$$X_i^- = x_i v_{2i+3}, \qquad X_i^+ = x_i v_{2i+5}.$$

For $0 \le j \le b-1$, add the two edges

$$Y_j^- = y_j v_{s+2j-1}, \qquad Y_j^+ = y_j v_{s+2j+1}.$$

A new vertex together with these two incident edges will be called a two-edge ear. The support of the ear x_i is the short cycle interval

$$v_{2i+3},\ v_{2i+4},\ v_{2i+5},$$

and the support of the ear y_j is

$$v_{s+2j-1},\ v_{s+2j},\ v_{s+2j+1}.$$

The x-ear region is the union of the supports of the x_i ears, namely

$$R_X = \{v_3, v_4, \ldots, v_{2a+5}\} \qquad (a \ge 1),$$

with $R_X = \varnothing$ when $a = 0$. The y-ear region is

$$R_Y = \{v_{s-1}, v_s, \ldots, v_{s+2b-1}\}.$$

The chord region is the small set

$$R_0 = \{v_{L-1}, v_0, v_1, v_2\}.$$

For later summations we also put

$$\begin{aligned} \mathcal{C} &= \{C_0, C_1, \ldots, C_{L-1}\}, & \mathcal{A} &= \{A, B\}, \\ \mathcal{X} &= \{X_i^-, X_i^+ : 0 \le i \le a-1\}, & \mathcal{Y} &= \{Y_j^-, Y_j^+ : 0 \le j \le b-1\}. \end{aligned}$$

Thus $E(G_{a,b}(L)) = \mathcal{C} \cup \mathcal{A} \cup \mathcal{X} \cup \mathcal{Y}$, a disjoint union. Figure 1 gives a schematic picture. The picture is not meant to be metrically accurate: the dotted arcs represent long intervals of ordinary cycle edges.

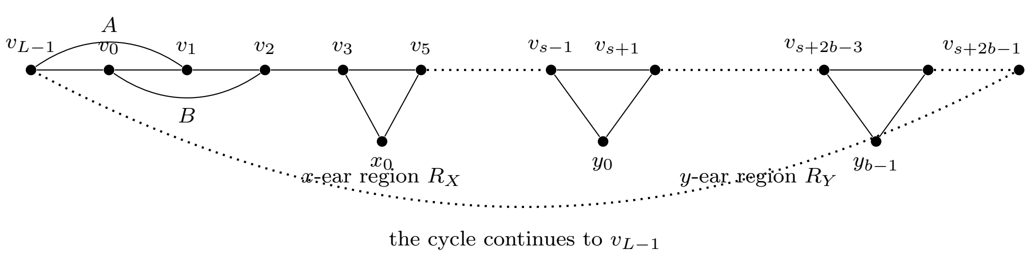

Figure 1: Schematic construction of $G_{a,b}(L)$. The graph consists of a long cycle, two short chords A and B, and two separated ear regions.

Lemma 1. *For every choice of $a \ge 0$ and $b \ge 1$ for which the displayed vertices are well-defined and the maximum degree is at most four, the graph $G_{a,b}(L)$ satisfies*

$$c(G_{a,b}(L)) = a + b + 3.$$

In particular, in the separated construction used for $c \ge 7$ below, the condition $L \ge 8c$ guarantees that $G_{a,b}(L)$ is a molecular graph.

Proof. The graph has L original cycle vertices and $a+b$ new ear vertices, hence

$$|V(G_{a,b}(L))| = L + a + b.$$

It has L cycle edges, two chords, and $2a+2b$ ear edges, hence

$$|E(G_{a,b}(L))| = L + 2 + 2a + 2b.$$

Therefore

$$|E| - |V| + 1 = (L + 2 + 2a + 2b) - (L + a + b) + 1 = a + b + 3.$$

It remains only to justify the molecular condition in the separated case. In the construction for $c \geq 7$ we choose

$$a = \left\lfloor \frac{c}{2} \right\rfloor - 2, \qquad b = c - 3 - a,$$

so $a + b + 3 = c$. We also impose $L \geq 8c$. Then

$$2a + 5 \leq c + 1 < s - 1,$$

because $s = \lfloor L/2 \rfloor \geq 4c$, and

$$s + 2b - 1 \leq s + c - 2 < L - 1.$$

Thus R_X, R_Y, and the chord region R_0 are pairwise separated along the cycle. The indices of the y-ears do not wrap around the cycle, and no y-ear endpoint lies in the chord region. Inside an ear region, consecutive ears may share one cycle vertex, but no cycle vertex is incident with more than two added ear edges. Since each cycle vertex already has degree two on C_L, every cycle vertex has degree at most four. Each new ear vertex has degree two. Hence $\Delta(G_{a,b}(L)) \leq 4$, and the graph is simple, connected, and molecular. □

3 The edge-distance count

For an edge $e = uv$ and a graph G, define

$$E_u(e) = \{f \in E(G) : \operatorname{dist}_G(u, f) < \operatorname{dist}_G(v, f)\},$$

$$E_v(e) = \{f \in E(G) : \operatorname{dist}_G(v, f) < \operatorname{dist}_G(u, f)\},$$

and

$$E_0(e) = \{f \in E(G) : \operatorname{dist}_G(u, f) = \operatorname{dist}_G(v, f)\}.$$

Thus

$$m_u(e \mid G) = |E_u(e)|, \qquad m_v(e \mid G) = |E_v(e)|, \qquad \mu_G(e) = \big||E_u(e)| - |E_v(e)|\big|.$$

The set $E_0(e)$ is the midpoint set of e. In a cycle-like graph it plays the role of a midpoint cut: when the cycle edges are read in their cyclic order, the inequality

$$\operatorname{dist}_G(u, f) < \operatorname{dist}_G(v, f)$$

can change only at an edge of $E_0(e)$ or next to a local shortcut, namely a chord or an ear support. This is why the calculation below splits into a finite local part and two long ordinary arcs.

The following lists constitute the distance-counting lemma used in the proof. For each displayed case, every edge of $G_c(L)$ occurs exactly once in the corresponding cycle or non-cycle list, and the listed integer is its exact contribution $\mu(e)$. The verification is mechanical from the definitions: for each oriented edge $e = uv$, compare

$$d_u(f) = \min\{\mathrm{dist}_G(u,r), \mathrm{dist}_G(u,t)\}, \qquad d_v(f) = \min\{\mathrm{dist}_G(v,r), \mathrm{dist}_G(v,t)\}$$

for every edge $f = rt$. On maximal ordinary cycle intervals outside $R_0 \cup R_X \cup R_Y$, the sign of $d_u(f) - d_v(f)$ changes only at a midpoint; in the finite shortcut regions the comparisons are made edge by edge. The separation guaranteed by $L \geq 8c$ ensures that these local comparisons do not interfere with one another.

3.1 General odd cyclomatic number

Assume first that $c \geq 7$ is odd. Write

$$c = 2a + 5, \qquad b = a + 2,$$

so $a = (c-5)/2$ and $b = (c-1)/2$. Let $G_c(L) = G_{a,b}(L)$ and assume $L \geq 8c$.

Cycle edges when $L = 2s$. The values of $\mu(C_i)$ are as follows:

$$\begin{aligned}
&\mu(C_i) = 0, \quad i \in \{0, 1, s+1\},\\
&\mu(C_i) = 1, \quad i = 2, \quad 2a+3 \leq i \leq s-2, \quad s+2b-1 \leq i \leq L-2,\\
&\mu(C_i) = 2, \quad 3 \leq i \leq 2a+2, \quad i = s-1, \quad s+2 \leq i \leq s+2b-2, \quad i = L-1,\\
&\mu(C_i) = 3, \quad i = s.
\end{aligned}$$

These four lines cover all L cycle edges. Therefore the cycle-edge contribution is

$$0 \cdot 3 + 1 \cdot (L - 2c + 3) + 2 \cdot (2c - 7) + 3 \cdot 1 = L + 2c - 8.$$

Non-cycle edges when $L = 2s$. The chord contributions are

$$\mu(A) = 2, \qquad \mu(B) = 0.$$

For every $0 \leq i \leq a-1$,

$$\mu(X_i^-) = \mu(X_i^+) = 2.$$

For the y-ears,

$$\mu(Y_0^-) = 2, \qquad \mu(Y_0^+) = 3, \qquad \mu(Y_1^-) = 0,$$

and all remaining Y-ear edges have contribution 2. Hence

$$\sum_{e \in \mathcal{A}} \mu(e) = 2, \qquad \sum_{e \in \mathcal{X}} \mu(e) = 2c - 10, \qquad \sum_{e \in \mathcal{Y}} \mu(e) = 2c - 3.$$

Combining the cycle and non-cycle contributions gives

$$\mathrm{Mo}_e(G_c(L)) = L + 6c - 19 \qquad (c \geq 7 \text{ odd},\ L \text{ even}).$$

Cycle edges when $L = 2s + 1$. The values of $\mu(C_i)$ are:

$$\begin{aligned}
\mu(C_i) = 0, \quad & i \in \{0, 2, L-2\},\\
\mu(C_i) = 1, \quad & i = 1, \quad 3 \leq i \leq 2a+2 \text{ with } i \text{ odd}, \quad 2a+3 \leq i \leq s-2,\\
& i = s, \quad s+1, \quad s+3 \leq i \leq s+2b-3 \text{ with } i-s \text{ even},\\
& s+2b-2 \leq i \leq L-3, \quad i = L-1,\\
\mu(C_i) = 2, \quad & i \in \{s-1, s+2\},\\
\mu(C_i) = 3, \quad & 3 \leq i \leq 2a+2 \text{ with } i \text{ even},\\
& s+3 \leq i \leq s+2b-3 \text{ with } i-s \text{ odd}.
\end{aligned}$$

Thus

$$\sum_{i=0}^{L-1} \mu(C_i) = L + 2c - 11.$$

Non-cycle edges when $L = 2s + 1$. Here

$$\mu(A) = \mu(B) = 1,$$

and for $0 \le i \le a - 1$,

$$\mu(X_i^-) = 1, \qquad \mu(X_i^+) = 3.$$

For the y-ears,

$$\mu(Y_0^-) = 2, \qquad \mu(Y_1^+) = 2,$$

$$\mu(Y_0^+) = \mu(Y_1^-) = 1, \qquad \mu(Y_j^+) = 1 \quad (2 \le j \le b - 1),$$

and

$$\mu(Y_j^-) = 3 \quad (2 \le j \le b - 1).$$

Consequently

$$\sum_{e \in \mathcal{A}} \mu(e) = 2, \qquad \sum_{e \in \mathcal{X}} \mu(e) = 2c - 10, \qquad \sum_{e \in \mathcal{Y}} \mu(e) = 2c - 4,$$

and therefore

$$\mathrm{Mo}_e(G_c(L)) = L + 6c - 23 \qquad (c \ge 7 \text{ odd},\ L \text{ odd}).$$

3.2 General even cyclomatic number

Assume now that $c \ge 8$ is even. Write

$$c = 2a + 4, \qquad b = a + 1,$$

so $a = (c - 4)/2$ and $b = (c - 2)/2$. Again let $G_c(L) = G_{a,b}(L)$ and assume $L \ge 8c$.

Cycle edges when $L = 2s$. The values of $\mu(C_i)$ are:

$$\begin{aligned}
\mu(C_i) = 0,\ & i = 0, \quad 3 \le i \le 2a + 1 \text{ with } i \text{ odd}, \quad i = s + 2, \\
& s + 4 \le i \le s + 2b - 2 \text{ with } i - s \text{ even}, \quad i = L - 1, \\
\mu(C_i) = 1,\ & i = 2, \quad 2a + 3 \le i \le s - 2, \quad i = s, \quad s + 2b - 1 \le i \le L - 2, \\
\mu(C_i) = 2,\ & i \in \{1, 2a + 2, s + 1\}, \\
\mu(C_i) = 4,\ & 4 \le i \le 2a \text{ with } i \text{ even}, \quad i = s - 1, \\
& s + 3 \le i \le s + 2b - 3 \text{ with } i - s \text{ odd}.
\end{aligned}$$

Hence

$$\sum_{i=0}^{L-1} \mu(C_i) = L + 2c - 10.$$

Non-cycle edges when $L = 2s$. The chords satisfy

$$\mu(A) = 0, \qquad \mu(B) = 2.$$

For the x-ears,

$$\mu(X_i^-) = 0 \quad (0 \le i \le a - 1),$$

$$\mu(X_i^+) = 4 \quad (0 \le i \le a-2), \qquad \mu(X_{a-1}^+) = 2.$$

For the y-ears,

$$\mu(Y_0^+) = 1, \qquad \mu(Y_1^-) = 2, \qquad \mu(Y_j^+) = 0 \quad (1 \le j \le b-1),$$

and

$$\mu(Y_j^-) = 4 \quad (j = 0 \text{ or } 2 \le j \le b-1).$$

Thus

$$\sum_{e\in\mathcal{A}} \mu(e) = 2, \qquad \sum_{e\in\mathcal{X}} \mu(e) = 2c-10, \qquad \sum_{e\in\mathcal{Y}} \mu(e) = 2c-5,$$

and

$$\mathrm{Mo}_e(G_c(L)) = L + 6c - 23 \qquad (c \ge 8 \text{ even},\ L \text{ even}).$$

Cycle edges when $L = 2s+1$. The values of $\mu(C_i)$ are:

$$\begin{aligned}
\mu(C_i) = 0,&\quad i \in \{0, 2a+1, s+2, s+2b\},\\
\mu(C_i) = 1,&\quad i = 3, \quad 5 \le i \le 2a-1 \text{ with } i \text{ odd}, \quad 2a+3 \le i \le s-2,\\
&\quad i = s, \quad s+4 \le i \le s+2b-2 \text{ with } i-s \text{ even},\\
&\quad s+2b+1 \le i \le L-3, \quad i = L-1,\\
\mu(C_i) = 2,&\quad i \in \{2, 2a+2, s+2b-1, L-2\},\\
\mu(C_i) = 3,&\quad i \in \{1, s+1\},\\
\mu(C_i) = 4,&\quad i = s-1,\\
\mu(C_i) = 5,&\quad 4 \le i \le 2a \text{ with } i \text{ even},\\
&\quad s+3 \le i \le s+2b-3 \text{ with } i-s \text{ odd}.
\end{aligned}$$

Therefore

$$\sum_{i=0}^{L-1} \mu(C_i) = L + 4c - 17.$$

Non-cycle edges when $L = 2s+1$. The chord contributions are

$$\mu(A) = 1, \qquad \mu(B) = 3.$$

For the x-ears,

$$\begin{aligned}
\mu(X_i^-) &= 1 \quad (0 \le i \le a-2), \qquad \mu(X_{a-1}^-) = 0,\\
\mu(X_i^+) &= 5 \quad (0 \le i \le a-2), \qquad \mu(X_{a-1}^+) = 2.
\end{aligned}$$

For the y-ears,

$$\mu(Y_0^-) = 4, \qquad \mu(Y_0^+) = 1, \qquad \mu(Y_1^-) = 3, \qquad \mu(Y_1^+) = 0,$$

and for $2 \le j \le b-1$,

$$\mu(Y_j^-) = 5, \qquad \mu(Y_j^+) = 1.$$

Hence

$$\sum_{e\in\mathcal{A}} \mu(e) = 4, \qquad \sum_{e\in\mathcal{X}} \mu(e) = 3c-16, \qquad \sum_{e\in\mathcal{Y}} \mu(e) = 3c-10,$$

and

$$\mathrm{Mo}_e(G_c(L)) = L + 10c - 39 \qquad (c \ge 8 \text{ even},\ L \text{ odd}).$$

Remark 1. *The preceding four classifications also show why no contribution larger than* 5 *occurs in the general construction. Indeed, every edge of* $G_c(L)$ *is explicitly listed, and each listed value belongs to* $\{0, 1, 2, 3, 4, 5\}$. *The bound* 5 *is not assumed; it is a consequence of the complete distance classification.*

For quick reference, Table 1 summarizes the four sums just obtained.

Table 1: Summary of the edge-distance count for $c \geq 7$.

c	L	$\sum_{e\in\mathcal{C}} \mu(e)$	$\sum_{e\in\mathcal{A}} \mu(e)$	$\sum_{e\in\mathcal{X}} \mu(e) + \sum_{e\in\mathcal{Y}} \mu(e)$	$\mathrm{Mo}_e(G_c(L))$
odd	even	$L + 2c - 8$	2	$4c - 13$	$L + 6c - 19$
odd	odd	$L + 2c - 11$	2	$4c - 14$	$L + 6c - 23$
even	even	$L + 2c - 10$	2	$4c - 15$	$L + 6c - 23$
even	odd	$L + 4c - 17$	4	$6c - 26$	$L + 10c - 39$

4 Small cyclomatic numbers

The separated two-ear-region construction above proves the result for all $c \geq 7$. The cases $3 \leq c \leq 6$ are short enough to handle directly. We record the details because they are also useful checks on the notation.

4.1 The case $c = 3$

Let H_L be obtained from C_L by adding the chord $v_{L-1}v_1$ and one new vertex z adjacent to v_{s-1} and v_{s+1}, where $s = \lfloor L/2 \rfloor$. Then

$$|V(H_L)| = L + 1, \qquad |E(H_L)| = L + 3,$$

so $c(H_L) = 3$. The graph is molecular. Directly from the definition of $\mu(e)$ one obtains

$$\mathrm{Mo}_e(H_L) = \begin{cases} L + 8, & L \geq 6,\ L \text{ even}, \\ L + 12, & L \geq 7,\ L \text{ odd}. \end{cases}$$

For example, when $L = 2s$, the two cycle edges next to the chord region contribute s each, four edges contribute 2, and all other edges contribute 0; hence $2s + 8 = L + 8$. When $L = 2s + 1$, the same comparison gives two large contributions $s + 1$ and $s + 2$, two contributions 3, four contributions 1, and all other contributions 0, giving $2s + 13 = L + 12$.

4.2 The cases $c = 4, 5, 6$

For $4 \leq c \leq 6$ we use $G_{a,b}(L)$ with the values of (a, b) shown in Table 2. The table includes explicit lower bounds on L; these are the values for which the displayed formulas hold and the graph is molecular. The entry $c = 6$, $L = 11$ is added separately because it realizes the missing value 32.

To indicate how the entries of Table 2 were obtained, we give the complete contribution counts in Table 3. Here $N(t)$ denotes the number of edges whose contribution $\mu(e)$ equals t; blank entries are zero. For instance, for $c = 5$ and L even there are $L - 7$ edges of contribution 1, six edges of contribution 2, two edges of contribution 3, and the remaining five edges have contribution 0; hence $\mathrm{Mo}_e = (L - 7) + 12 + 6 = L + 11$.

Table 2: Small cyclomatic numbers.

c	(a,b)	Condition on L	$\mathrm{Mo}_e(G_{a,b}(L))$
4	$(0,1)$	$L \geq 6,\ L$ even	$L+9$
4	$(0,1)$	$L \geq 7,\ L$ odd	$L+11$
5	$(0,2)$	$L \geq 8,\ L$ even	$L+11$
5	$(0,2)$	$L \geq 9,\ L$ odd	$L+7$
6	$(1,2)$	$L \geq 14,\ L$ even	$L+13$
6	$(1,2)$	$L \geq 13,\ L$ odd	$L+21$
6	$(1,2)$	$L = 11$	32

Table 3: Contribution counts for $c = 4, 5, 6$.

c	parity of L	$N(0)$	$N(1)$	$N(2)$	$N(3)$	$N(4)$
4	even	3	$L-3$	2	2	
4	odd	2	$L-3$	3	2	
5	even	5	$L-7$	6	2	
5	odd	3	$L-1$	4		
6	even	7	$L-7$	6	2	
6	odd	6	$L-9$	5	4	2

5 Proof of the inverse theorem

Theorem 1. *For every fixed integer $c \geq 3$, there exists a finite set $B_c \subset \mathbb{N}$ such that every $p \in \mathbb{N} \setminus B_c$ is the edge Mostar index of a c-cyclic molecular graph.*

Proof. We prove a stronger explicit statement: for each fixed c, every integer above a stated threshold is attained.

For $c = 3$, use H_L. If $p \geq 18$ is even, choose $L = p - 8$. Then L is even and at least 10, so $\mathrm{Mo}_e(H_L) = L + 8 = p$. If $p \geq 19$ is odd, choose $L = p - 12$. Then L is odd and at least 7, so $\mathrm{Mo}_e(H_L) = L + 12 = p$. Hence every $p \geq 18$ is attained when $c = 3$.

For $c = 4$, use $G_{0,1}(L)$. If $p \geq 17$ is odd, choose $L = p - 9$, which is even and at least 8. If $p \geq 18$ is even, choose $L = p - 11$, which is odd and at least 7. Table 2 gives $\mathrm{Mo}_e = p$ in both cases. Thus every $p \geq 17$ is attained.

For $c = 5$, use $G_{0,2}(L)$. If $p \geq 19$ is odd, choose $L = p - 11$, which is even and at least 8. If $p \geq 20$ is even, choose $L = p - 7$, which is odd and at least 13. Again Table 2 gives $\mathrm{Mo}_e = p$. Thus every $p \geq 19$ is attained.

For $c = 6$, use $G_{1,2}(L)$. The value $p = 32$ is realized by $G_{1,2}(11)$. If $p \geq 33$ is odd, choose $L = p - 13$, which is even and at least 20. If $p \geq 34$ is even, choose $L = p - 21$, which is odd and at least 13. Therefore every $p \geq 32$ is attained.

Now assume $c \geq 7$ is odd. By the calculation summarized in Table 1,

$$\mathrm{Mo}_e(G_c(L)) = \begin{cases} L + 6c - 19, & L \text{ even}, \\ L + 6c - 23, & L \text{ odd}, \end{cases} \qquad (L \geq 8c).$$

Set

$$N_c = 14c - 19.$$

If $p \geq N_c$ is odd, choose

$$L = p - (6c - 19).$$

Since both p and $6c - 19$ are odd, L is even; moreover

$$L \geq (14c - 19) - (6c - 19) = 8c.$$

Thus $\mathrm{Mo}_e(G_c(L)) = p$. If $p \geq N_c$ is even, choose

$$L = p - (6c - 23).$$

Then L is odd and

$$L \geq (14c - 18) - (6c - 23) = 8c + 5 > 8c,$$

so again $\mathrm{Mo}_e(G_c(L)) = p$. Therefore every $p \geq 14c - 19$ is attained for odd $c \geq 7$.

Finally assume $c \geq 8$ is even. The calculation summarized in Table 1 gives

$$\mathrm{Mo}_e(G_c(L)) = \begin{cases} L + 6c - 23, & L \text{ even}, \\ L + 10c - 39, & L \text{ odd}, \end{cases} \qquad (L \geq 8c).$$

Set

$$N_c = 18c - 38.$$

If $p \geq N_c$ is even, choose

$$L = p - (10c - 39).$$

Then L is odd and

$$L \geq (18c - 38) - (10c - 39) = 8c + 1 > 8c,$$

so $\mathrm{Mo}_e(G_c(L)) = p$. If $p \geq N_c$ is odd, choose

$$L = p - (6c - 23).$$

Then L is even and

$$L \geq (18c - 37) - (6c - 23) = 12c - 14 \geq 8c.$$

Thus $\mathrm{Mo}_e(G_c(L)) = p$ in this case as well.

Consequently every fixed $c \geq 3$ has a finite exceptional set. For example, one may take

$$B_c = \begin{cases} \{1, 2, \dots, 17\}, & c = 3, \\ \{1, 2, \dots, 16\}, & c = 4, \\ \{1, 2, \dots, 18\}, & c = 5, \\ \{1, 2, \dots, 31\}, & c = 6, \\ \{1, 2, \dots, 14c - 20\}, & c \geq 7 \text{ odd}, \\ \{1, 2, \dots, 18c - 39\}, & c \geq 8 \text{ even}. \end{cases}$$

This proves the theorem. $\square$

Corollary 1. *Conjecture 4.2 of Alex and Gutman is true. More precisely, it is true in the stronger class of molecular graphs.*

6 Chemical and mathematical remarks

The proof uses only graphs of maximum degree at most four. Thus the constructions satisfy the usual valence restriction for carbon skeletons, although they are intended as graph-theoretic molecular graphs rather than as claims about the synthesis or stability of particular chemical compounds. The construction keeps the cyclomatic number fixed while increasing the length of a ring-like backbone. In chemical language, this separates ring complexity, encoded by c, from molecular size.

The exceptional sets displayed in Theorem 1 are not claimed to be minimal. Determining the exact forbidden values for each fixed $c \geq 3$ is a finer inverse problem. Another natural direction is to ask whether analogous inverse theorems hold for the total Mostar index and the total edge Mostar index, where the summation is over all unordered vertex pairs rather than only over adjacent pairs. The present proof relies on the bond-additive nature of Mo_e and does not directly settle those total variants.

Declarations

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